

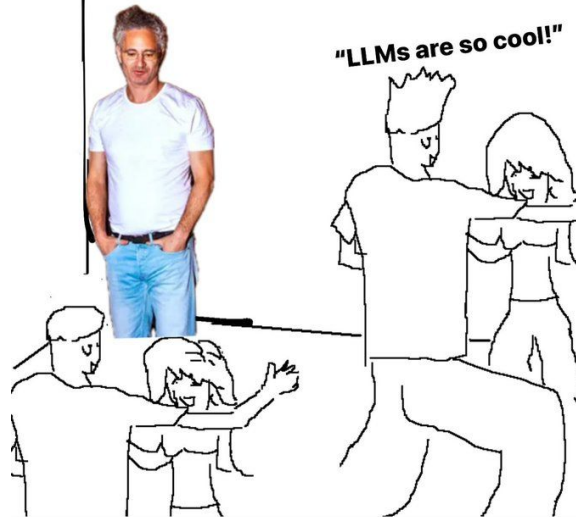
InfiniteCartridge

an endless gameboy generative machine

concept

- Handheld that never plays the same game twice
- Games are flashed and **run on device** (like cartridges in a Game Boy would)
- Each game is designed from scratch, never to be repeated by you or other players (think No Man's Sky planets)
- Three Anthropic agents:
Designer (Assets / Game design) + **Engineer** (Code-gen) + **Composer** (custom 8-bit soundtrack)
- Real 8-bit music on og **analog audio chip**
- Agents and flashing/audio
orchestrated by a console OS that runs **on the laptop**

**They don't know that
they need an ontology**



"LLMs are so cool!"

**"I loved your PowerPoint.
I want AI!"**

Ontology (game randomisation)

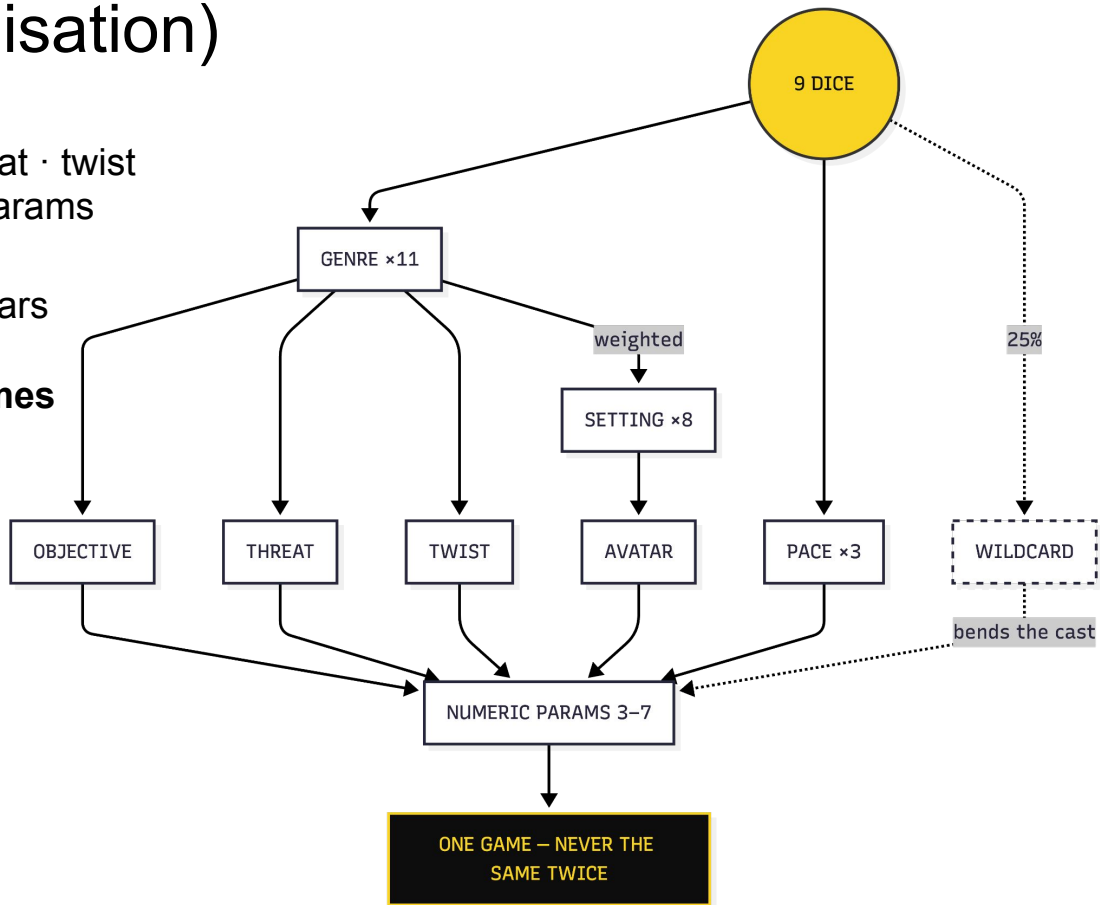
Every game is **9 dice**: genre · objective · threat · twist
· pace · setting · avatar · wildcard · numeric params

13,662 discrete skeletons × up to 10^7 param vars

≈ **billions of games**

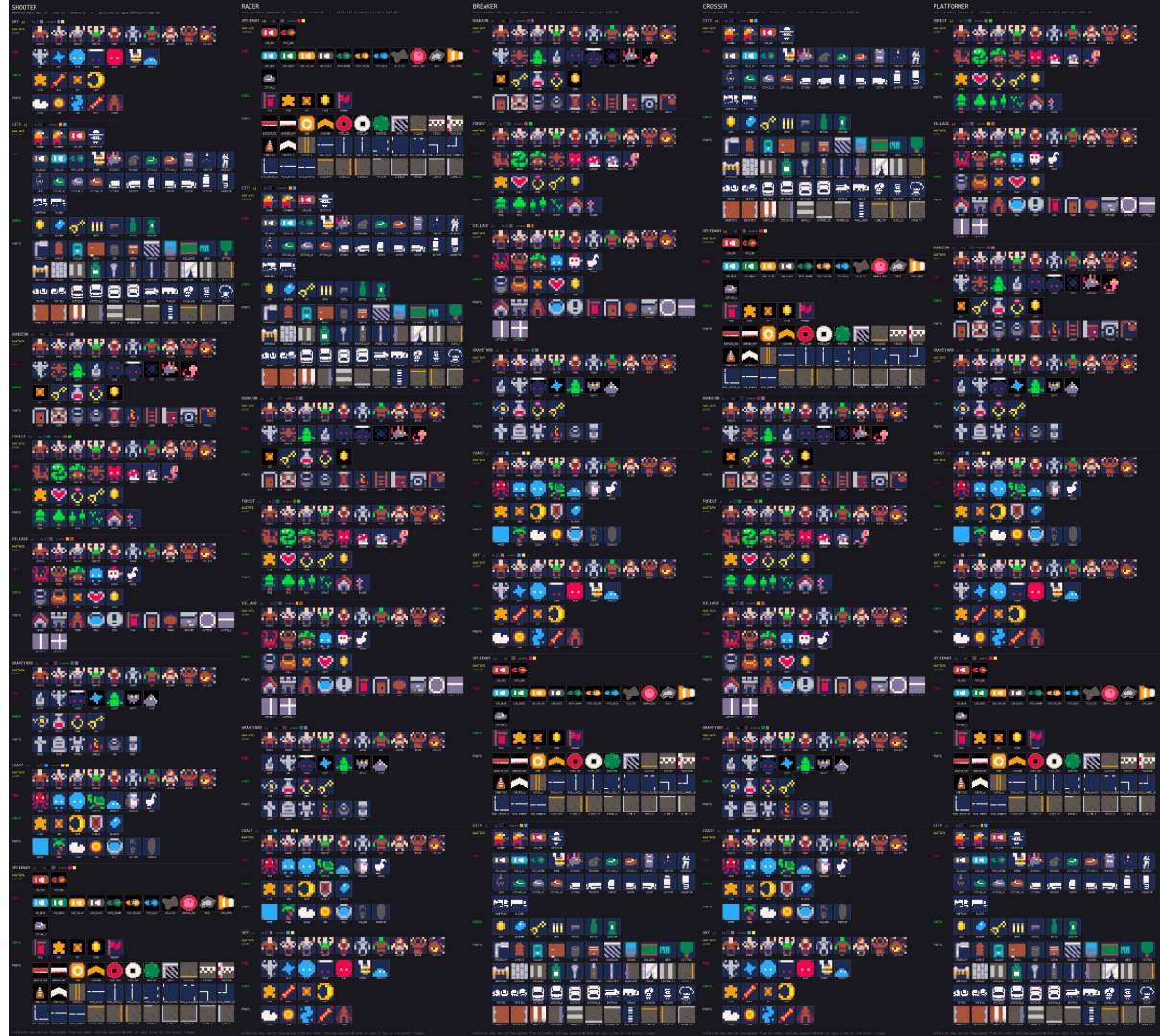
Anti-repeat memory for every die:
no genre twice in 4 games,
no repeat pairing in 8

Mercy is systemic: 10 lives, goals capped
by spawn economy.
every game winnable by construction
(or at least should be, ya'know :)



Agentique: Level Design

- 8x8px sprites (360 to choose from)
- Sorted by genre / setting and tagged
- **LLM picks the cast:**
character · enemies · textures · props
- **Designs the level as a 20×8 ASCII map:**
collisions, terrain, spawn points
- Applies textures by legend:
every map letter is a sprite



Agentique: code-gen + verification

- The C++ engine is fixed: screen, input, HUD, 30fps shell.
- Generated game code is ~150-200 lines ArduinoC sketch
- Verification loop:

Static check → arduino-cli compile → errors fed back, ≤3 repair rounds.

(compilermaxxing for unit tests)

- Anti-slop protection: delay / while / heap / raw screen access - all classic footguns are caught and vetoed.
- Pipelined: N+1 written while N compiles

Composing 8-bit music like your grandpa did it



- Every game gets own OST: composed while it compiles
- Target: **AY-3-8912** analog audio chip ('80 ZX Spectrum)
- The **agent composes sheet music** note-by-note:

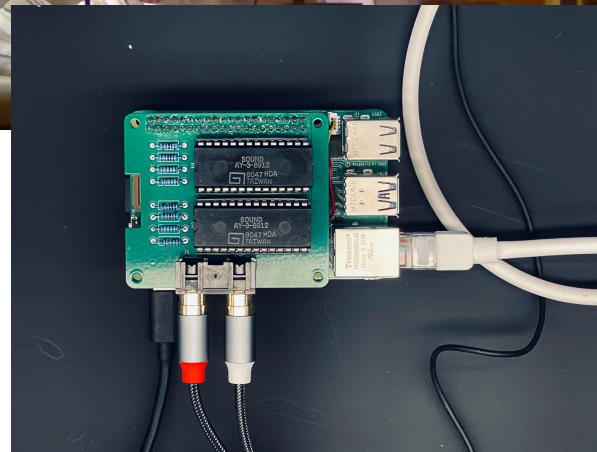
<https://github.com/gasman/papaya>

melody, bassline, harmony, drums

on a 16th-note grid, 3 voices

JSON -> **MIDI as output**

- A service in console OS converts to **.PT3**
 - RPi5 with the audio chip exposes an API
- the converted song is uploaded there
- The orchestrator plays it the moment the game flashes



[Demo]

Thank you!